

## Math 271: Mathematical Methods

### Class Test 3 — Grading Conventions

#### Grading conventions (applies to both questions)

- C1. **Choice of rubric** A question is graded under the *rubric*<sup>1</sup> that matches the proof the answer script actually presents. If an answer script mixes routes, it is graded under the rubric that is most favourable to the student, giving credit for each item that the answer script establishes.
- C2. **Other correct proofs.** Any complete and correct proof scores full marks, whether or not it follows one of the rubrics furnished here. For such an answer script, I have mapped its steps onto the closest rubric and awarded the corresponding items.
- C3. **Equal mathematics, equal marks.** The rubrics for the same question are deliberately itemised so that the same mathematics attracts the same total, whichever way it is written up.
- C4. **Deductions.** Deductions<sup>2</sup> are  $-2$  to  $-8$  marks in every rubric item.
- C5. **Use of Question 1 inside Question 2.** A student may quote the result of Question 1 in Question 2 even if Question 1 was not attempted, or was attempted unsuccessfully; no marks are transferred between the two questions.
- C6. **Ellipsis over a uniform step.** A recursive construction may be signalled by “and so on”, “continuing inductively”, “repeating this argument” or the like, and I accept this in full, *provided the step being repeated is uniform*: the step at stage  $k$  is the first step verbatim with  $n_1$  replaced by  $n_k$ , and the valid logical statement that permits the step holds at every stage. What makes such a repetition legitimate is that licensing fact — the valid logical statement — for instance, “no index  $n \geq n_1$  is a peak” . So where the first step is correct, the repetition is signalled, and the examinee has stated the valid licensing fact, I award the construction item *in full*, whether or not the generic clause  $n_k \mapsto n_{k+1}$  is written out. An ellipsis (...) is fatal only where the step is *not* uniform, i.e. where the sentence one would have to repeat is not available at the second stage; there the construction is not a proof, and I say so, with the reasoning, in the rubric concerned.
- C7. **Unestablished hypothesis.** An item whose award depends on a statement the answer script asserts but does not establish is worth *at most* half of the marks it would otherwise attract — even where the missing statement is true, and even where the ingredients for proving it appear elsewhere on the page, but not connected with the item in question.
- C8. **False statement.** A statement that is false as written voids nothing by itself, but every item resting on it is affected, and a deduction of at least  $-2$  is charged once per question under C4. This is the convention that distinguishes a step that is *missing* from a step that is *wrong*.
- C9. **Hypotheses of a quoted theorem.** A named theorem whose hypothesis is not verified attracts the parial credit and no more (16 of 22 for Bolzano–Weierstrass; 10 of 16 for the Monotone Convergence Theorem where only one direction is treated). If the theorem is also misstated, half of the partial credit.

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<sup>1</sup>A *rubric* here means a way of answering a particular question - as you know already, there are actually many ways to answer the same question

<sup>2</sup>Deduction should be understood differently than the absence of any valid content to mark against

- C10. **Absent means zero.** No credit for a definition that is used but not stated, none for asserting the conclusion of an item, and none for an intention that is not written down. Quoting a result the paper allows the student to quote (Question 1 inside Question 2, Hints 1 and 2) remains fully creditable: that is C6, and C11 does not touch it.
- C11. **Case analysis that does not help.** 5 points are awarded for the two exhaustive cases in the proofs. These points are awarded for recognizing the correct splitting of the cases that the proof actually runs on. Where two cases are on the page but they are not the valid split, or they are not exhaustive, or one of them is false, the item scores 0; the partial award of 3-marks survives only for the right split stated imprecisely. A case analysis that is unhelpful or wrong is not partial progress towards the proof.